

CHUANG, CHENG-SHUN (RYAN)

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PROFESSIONAL SUMMARY

Computer Science M.Eng. student and Graduate Research Assistant focused on health AI, computer vision, full-stack research platforms, and cloud deployment. Experience building research software with React, TypeScript, Python, Flask/FastAPI, PostgreSQL, SQLAlchemy, PyTorch, AWS, GCP, Docker, Terraform, and GitHub Actions.

EDUCATION

Virginia Tech · Alexandria, VA, USA

Jan. 2025 – Jan. 2027 (Expected)

Master of Engineering in Computer Science

- GPA: 4.0/4.0
- Relevant coursework: Machine Learning, NLP, AI Tools for Software Engineering, Advanced Topics in Data & Information Systems, Advanced Topics in Systems, Verification & Validation of Scientific Computing, Research & Thesis

National Taipei University (NTPU) · New Taipei City, Taiwan

Sep. 2019 – June 2024

Bachelor of Arts in Economics

- Overall GPA: 3.95/4.0 | Last 60 credits: 4.0/4.0 | CS Courses: 4.0/4.0
- Coursework: OS, Databases, ANN & Deep Learning, Computer Vision
- Awards: Academic Excellence (3×), Falcon Award, Dean's Award

PUBLICATIONS

- Cusati, J., Chuang, C.-S. **Virginia Tech Transportation Safety Index (VTTSI)**. *arXiv:2606.21154 [cs.CY]*, 2026. [Second Author] Proposed a real-time, cloud-native transportation safety index combining connected-vehicle telemetry, Empirical Bayes crash stabilization, and CRITIC-weighted MCDM scoring.
- Chuang, C.-S. et al. **Impact of Linear Dimensionality Reduction on Machine Learning Performance: A Case Study on Fashion-MNIST**. *Computing Conference 2026*, In Press. [Main Author] Evaluated how PCA, LDA, Factor Analysis, and Random Forest feature selection affect downstream classifiers; SVM+PCA achieved top accuracy of 0.8995.
- [Co-Author] **Design of a Low-Cost, High-Resolution Underwater Acoustic Imager with Integrated Deep Learning Enhancement**. *IEEE Sensors Journal*, 2025. Integrated a lightweight U-Net (~1M params) achieving 7–9× axial and 2–3× lateral resolution improvement over baseline ASAI.

RESEARCH PROJECTS

Virginia Tech Transportation Safety Index (VTTSI)

Sep. 2025 – Dec. 2025

Python · Empirical Bayes · MCDM (CRITIC/SAW/EDAS/CODAS) · FastAPI · Streamlit · GCP

- Developed a real-time road safety scoring system (0–100) integrating VDOT crash history with connected-vehicle telemetry at 15-minute resolution.
- Published the framework as an arXiv preprint: arXiv:2606.21154.
- Applied Empirical Bayes crash modeling with uplift factors (speed deficit, VRU conflicts) and a CRITIC-weighted hybrid MCDM framework (SAW–EDAS–CODAS) for multi-criteria ranking.

Taiwanese Car Plate Recognition

May 2023 – June 2023

Python · YOLOv8 · Computer Vision · Object Detection

- Trained YOLOv8 models to recognize Taiwanese license plates from moving-vehicle video; achieved **mAP@IoU = 0.5 of 0.986** (detection) and 0.966 (OCR), with zero errors in static parking-lot scenarios.

EXPERIENCE

Graduate Research Assistant — ResonaCare and ETV Surgical Video AI

May 2026 – Present

Virginia Tech

- Building ResonaCare, a React/TypeScript and Flask/PostgreSQL surgical-video rating platform with rater, researcher, and admin workflows; implemented assignment-scoped 3–30s segment annotation, OSATS category ratings, review/invalidation, CSV exports, and consent-gated study-event logging.
- Integrated AWS deployment and AI evidence paths: S3 media upload, Amplify/Lambda/RDS stack, Bedrock voice classification, SageMaker/ResNet-18 frame analysis, and GitHub Actions CD.
- Conducting Endoscopic Third Ventriculostomy (ETV) surgical-video instance segmentation research across 11 anatomical/tool classes; benchmarking YOLOv8, YOLO11, YOLO12, and YOLO26 on 15,026 annotated frames from 63 surgeries with real-time Tesla T4 deployment constraints.

TVBS News — QA Engineer Intern

Jan. 2024 – June 2024

Taipei, Taiwan

- Ensured mobile app quality through systematic bug identification and regression testing.
- Built and tested RESTful APIs using MVC architecture, improving data retrieval speed and stability.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Machine Learning and Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, YOLOv8, U-Net, ResNet-18

Software Engineering: React, Flask, FastAPI, SQLAlchemy, Docker, Terraform, GitHub Actions, Linux, WSL2

Cloud and Databases: AWS Amplify, AWS Lambda, Amazon RDS, Amazon S3, Amazon SageMaker, Amazon Bedrock, GCP, PostgreSQL, MySQL, Cloud SQL